GRE Quantitative Reasoning

모의고사 Set 13 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 13 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Section 1 12 Questions · 21 Minutes

Question 1 대소비교

1127. The leg of isosceles triangle A is 8, while the leg of isosceles triangle B is 10.
Quantity A: The area of isosceles triangle A
Quantity B:
The area of isosceles triangle B

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

795. $1 < x+y$
Quantity A: x^2+y^2
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

700. n is a positive integer.
Quantity A: The remainder when $5(10^n) + 1$ is divided by 3
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

998. n is a positive integer.
Quantity A: The remainder when $3^{4n+2}+5$ is divided by 10
Quantity B: 4

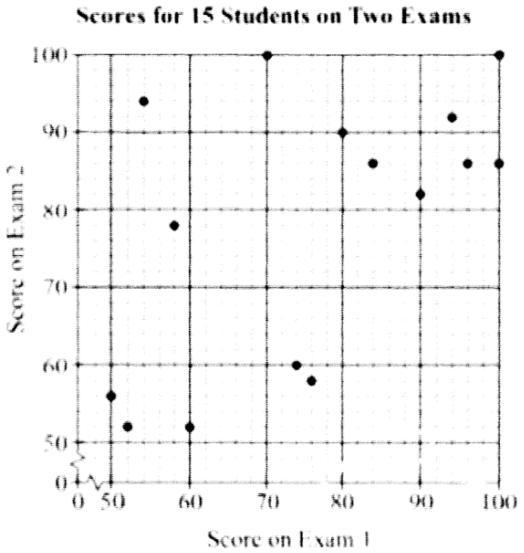
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

974. A coin collector bought a certain coin for $25, sold it for $35, then bought is back for $45, and finally sold it for $55. What was the gain or loss for the collector on this series of transactions?
A$30 gain
B$20 gain
C$10 gain
D$10 loss
ENeither gain nor loss

A B C D E

Questions 6 to 8 are based on the following data.



Note: The two coordinates of each dot on the graph represent the two exam scores of a student, where each shore is a whole number.

Question 6 객관식

576. The median of the scores on exam 1 was how much higher or lower than the median of the scores on exam 2?
A10 lower
B6 lower
C2 higher
D4 higher
E8 higher

A B C D E

Question 7 객관식

577. One of the 15 students had a total score of 182 on the two exams. What is the weighted mean of that student's two scores if the score on exam 1 has a weight of 40 percent and the score on exam 2 has a weight of 60 percent?
A89
B90
C91
D92
E93

A B C D E

Question 8 객관식

578. A pair of students is to be chosen to write a certain report, and each of the chosen students must have a total score of 170 or higher on the two exams. How many different pairs of students meet this requirement?
A6
B8
C16
D21
E28

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 객관식

1182. It costs d dollars to buy n nails. At this rate, what is the cost of n+1,850 nails, in dollars?
A. (1850nd+d)/n
B. (nd+d)/1850n
C. (1850d+n)/d
D. (1850n+n2)/d
E. (1850d+nd)/n

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

1000. A lecture hall has 15 rows of seats. There are n-2 seats in the first row and n seats in each of the other rows. ff there are no other seats in the lecture hall and the total number of seats in the lecture hall is between 180 and 200, what is the total number of seats in the lecture hall?

Question 11 복수선택

820. A furniture store purchased sofas for a wholesale price of $400 each and sold most of them for a retail price of $560 each. In a summer sale, the retail price of the last few sofas was reduced by x dollars, and they were sold at this reduced price. The store's profit from selling the last sofa was more than 15 percent of its wholesale price. Which of the following could be the value of x? Indicate all such values.
A50
B75
C100
D125

Indicate all that apply. A B C D

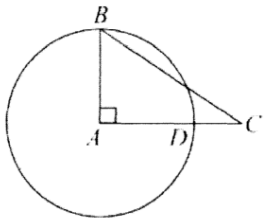
Question 12 객관식

976. If r and s are positive integers, which of the following is equal to r percent of s percent of 200?
A (rs)/100
B (rs)/50
C rs
D 2rs
E 200rs

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Section 2 15 Questions · 26 Minutes

Question 1 대소비교



494.In the figure, point A is the center of the circle and points B and D lies on the circle. The length of DC is one-half of the length of AD.
Quantity A: The area of sector ABD
Quantity B: The area of triangle ABC

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

959. a>b>0, r>s>0
In the xy-plane, line l crosses the y-axis at (0,a) and crosses the x-axis at (r,0), while line m crosses the y-axis at (0,b) and crosses the x-axis at (s,0).
Quantity A: The slope of line l
Quantity B: The slope of line m

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

457. n is a positive integer, x is the units digit of 7^n, y is the units digit of 3^n
Quantity A: |x-y|
Quantity B: 3

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1137. The median of the five measurements in set X is 65 and the median of the five measurements in set Y is 75. All ten measurements in sets X and Y are between 60 and 80.
Quantity A: The median of the ten measurements in sets X and Y combined
Quantity B: 70

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

536. In a probability experiment, R, S, and T are events such that P(S)=P(T)=x, P(R)=kx, and P(R or S) < P(S or T), where k and x are positive numbers. Events R and S are mutually exclusive, and events S and T are independent.
Quantity A: k
Quantity B: 1-x

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

848. If $-1<y<0$, which of the following must be true?
A $y < y^2 < y^3 < y^4$
B $y^3 < y < y^2 < y^4$
C $y^3 < y < y^4 < y^2$
D $y < y^3 < y^4 < y^2$
E $y < y^3 < y^2 < y^4$

(A) (B) (C) (D) (E)

Question 7 객관식

546. The phone extensions of a certain company run consecutively from 3000 to 3799. The phone extensions that began with the digits 33 or 34 belong to Division 1 of the company. The phone extensions that end with the digit 0 or 1 belong to managers. What percent of the phone extensions of the company belong to managers in Division 1?
A3%
B4%
C5%
D6%
E7%

(A) (B) (C) (D) (E)

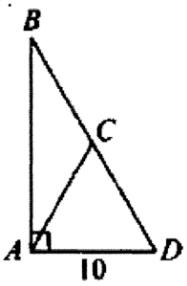
Question 8 객관식

1017. How many ways can the integers 3, 4, 6, 7, and 8 be arranged using each digit only once to form a five-digit integer such that the middle digit is odd?
A24
B32
C48
D64
E72

A B C D E

Question 9 복수선택

1177.



In the figure above, triangle ACD is equilateral. Which of the following lengths can be determined from the information given? Indicate all such lengths.
A.AB
B.AC
C.BC
D.CD

Indicate all that apply. A B C D

Question 10 주관식

696. Rodrigo's locker number has 3 different digits, the sum of which is 12. The sum of any two digits in the number is less than 10, and the digits are in decreasing order from left to right. What is Rodrigo's locker number?

Question 11 객관식

445. P, Q, and R are three points in a plane that are not all on the same line. Which of the following describes the set of all points in the plane that are equally distant from points P, Q and R?
AA circle
BA point
CA triangle
DTwo lines
EThree lines

A B C D E

Question 12 객관식

1204. At a constant rate of x/6 miles for every y seconds, how many minutes would it take an aircraft to travel z miles? (xyz≠0)
A.60yz/x
B.10yz/x
C.yz/60x
D.yz/10x
E.xyz/60

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 13 복수선택

366. If p is an prime number greater than 5,and if 5 is a factor of $p+p^2$,which of the following might be the remainder when p is divided by 5? Indicate all such numbers.
A0
B1
C2
D3
E4

Indicate all that apply. [A] [B] [C] [D] [E]

Question 14 객관식

896. If twice the sum of the positive integers x, y, and z is divided by 7, the remainder is 1. What is the remainder if the sum of x, y, and z is divided by 7?
A0
B1
C2
D3
E4

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 15 객관식

458. R and t are positive integers If r=50-t, then what could be the greatest possible median of r, t and 28?
A23
B24
C26
D28
E29

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ